



Study on the *cold* and *hot* properties of medicinal herbs by thermotropism in mice behavior[☆]

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ABSTRACT

It is a common sense that chewing a mint leaf causes a *cold* feeling, while masticating a piece of ginger root is associated with a *hot* sensation. The Traditional Chinese Medicine has termed this phenomenon as *cold* and *hot* properties of herbs and applied them in treating certain human diseases successfully for thousands of years. Here, we have developed an Animal Thermotropism Behavior Surveillance System, and by using this device and other approaches, we not only verified the existence of, but also characterized and quantitated the *cold* and *hot* properties of medicinal herbs in animal behavioral experiments. The results suggested that the *hot* and *cold* properties of herbal drugs indeed correlated with the alteration of animal behavior in search for residence temperature.

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1. Introduction

The *cold* or *hot* properties of herbal drugs and the application of corresponding knowledge in diagnosis, differentiation and treatment of diseases constitute an important aspect in Traditional Chinese Medicine (TCM) (Lei and Zhang, 1998; Gao, 1992; Xiao, 2008; Zhou et al., 2004). However, the so-called *cold* and *hot* properties of herbal drugs are a highly abstracted theory and is, thus far, difficult to find a scientific way to prove and characterize it. If these properties cannot be characterized and quantified, how to ascertain the objective existence of the *cold* or *hot* properties of herbal drugs? Furthermore, in absence of experimental evidence of these *cold* or *hot* properties, how to confirm the scientific nature of the theory?

The *cold* and *hot* properties of medicinal herbs have been plaguing scientists of modern medical scientists. Since 1970s scientists from China, Japan, and Korea have been applying physiological and pharmacological approaches to investigate the endocrinological (Ma, 1984), immunological (Qin et al., 2005) and neurological (Li et al., 1999; Fu and Guo, 1996) changes associ-

ated to the *cold* and *hot* properties of herbal drugs, attempting to unlock the myth of the ancient theory. Many findings (indicators) have been documented, such as enhanced or weakened certain neurological (sympathetic/parasympathetic) functions, up- or down-regulated neurotransmitter molecules, altered the ratio of cyclic adenosine monophosphate to cyclic guanosine monophosphate (cAMP/cGMP) (Fu et al., 1994), etc. Unfortunately, the results of these studies are generally non-convincing and less-conclusive, and by no means can represent the macro, whole-bodily, and dynamic nature of *cold* and *hot* properties of herbal drugs. The indicators found in these investigations may be relevant to, but in no way should be taken equal to the *cold* and *hot* properties of medicinal herbs.

According to established TCM theory, *cold* or *hot* properties of herbal drugs were emphasized by the relationship between environment temperature and the body feelings to the environment. In a cold environment, the body feels *cold* and tends to move to a relatively warm environment in order to survive (Gordon et al., 1998, 2000; Campbell, 2008; Liu et al., 2004; Wang et al., 2005), showing the tropism for higher-temperature; however, the intervention using drugs with *hot* properties helps our body increase the tolerance to the cold environment, showing different temperature tropism. The ancient theory of TCM, *to treat cold with hot* and *to treat hot with cold*, was the primitive awareness of such phenomenon. In this report, we studied the *cold* and *hot* properties of medicinal herbs from a macro and whole-bodily view by observing the drugs' impacts on animal behaviors and functions (Andersen et

[☆] Note: The Animal Thermotropism Behavior Surveillance System has been granted a Chinese patent # ZL200820004444.2. U.S. patent is on pending.

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al., 2006), aiming to establish an objective evaluation method for determining the cold and hot properties of herbal drugs.

2. Materials and methods

2.1. Herbs and assay kits

Rhubarb (*Radix rhei*), Coptidis (*Rhizoma coptidis*), Strobal (*Radix aconiti Lateralis praeparata*), Ginger (*Rhizoma zingiberis*), Ginseng (*Radix ginseng* processed with steam) and Gypsum (*Gypsum fibrosum*) were bought from Beijing Tong-ren-tang Herbal Drug Store. All of them were identified by Prof. Xiao-he Xiao, a taxonomist at China Military Institute of Chinese Material Medica. The voucher specimens of these herbs were deposited in the institute. The herbs were grinded and decocted twice with water for 30 min each. The resulting extracts were decanted, filtered and evaporated to dryness under reduced pressure. This dried extract was re-dissolved and dispersed in warm water and then cooled for intragastric administration to mice.

Adenosine triphosphatase assay kit (Lot. 20081113), coomassie brilliant blue protein determination kit (Lot. 20081113), superoxide dismutase assay kit (Lot. 20081113), succinate dehydrogenase assay kit (Lot. 20081113), malondialdehyde assay kit (Lot. 20081113) and total antioxidative capacity (Lot. 20081113) were purchased from the Nanjing Institute of Biological Engineering.

2.2. The animal thermotropism behavior surveillance system

In order to monitor the thermotropism behavior of animals, we have developed a device termed “Intelligent Monitoring System for Animal Thermotropism Behaviors”. The system consists of three major parts, namely, the automatic temperature controlling

unit (Fig. 1a), remote monitoring unit (Fig. 1b), and data processing unit (Fig. 1c). By comparing the behavior differences between the normal animals and the drug-treated animals in search for own most-suitable resident temperature (either on tolerable high-temperature, optimal-temperature, or tolerable low-temperature plate), the impacts of cold and hot properties of herbal drugs on animals were then determined. The automatic temperature controlling unit controls two isolated but thermostatic plates, and across these plates there are six channels for animal free migration (Fig. 2a). The device can survey the movements of six animals at same time, thus increases the repeatability and compatibility of data. Experimental animals were put on the temperature-controlling plate, and parameters such as the respective resident duration (s) of animals in different zones, the times of movement, can be observed, recorded, and documented in real-time through visual technology, i.e., surveillance using video camera, followed by real-time computer processing of artificial intelligence (Fig. 2b). The positional coordinates (X, Y) of an animal was determined in each frame of image, and the trajectories of animals were acquired by joining of the coordinates of each animal in all frames $Y_t = f(X_t)$, followed by computer processing of the duration and frequency of each animal in different zones (Fig. 2c and d). Video recognition software was based on Paulo's algorithms (Paulo et al., 2007), and data analysis software was written using Visual Basic 6.0 language.

2.3. Animals and drug administration

Male Kunming mice (license no.: SCXK2000-0010) of approximately 18 g were obtained from National Institute for the Control of Pharmaceutical and Biological Products of China. The mice were randomly divided into groups and administered intragastrically with the herbal extracts from Rhubarb (*Radix rhei*) 20 g kg⁻¹, Strobal

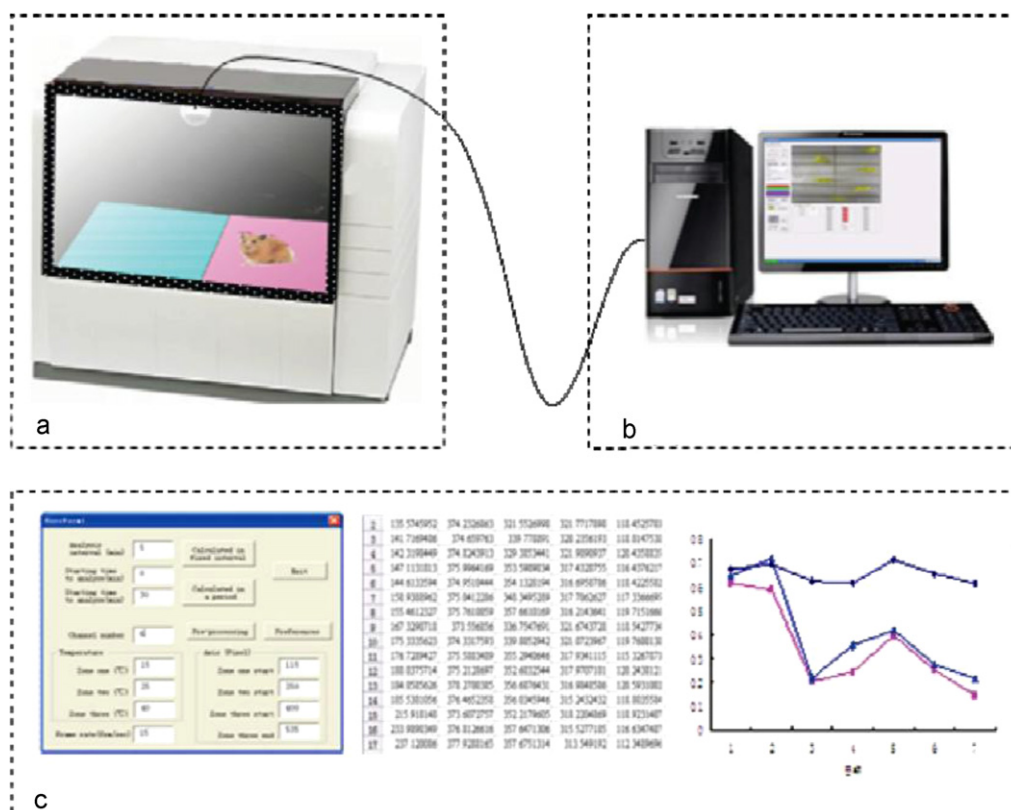


Fig. 1. The Animal Thermotropism Behavior Surveillance System which consists of three major parts. (a) Temperature controlling unit which contains two different temperature zones, and six channels across these zones for mice to move freely to search for their best-livable temperature. (b) A remote monitoring unit connecting to the temperature controlling unit monitors and records the animal movement. (c) Data processing unit which analyses the collected animal data and documents the results.

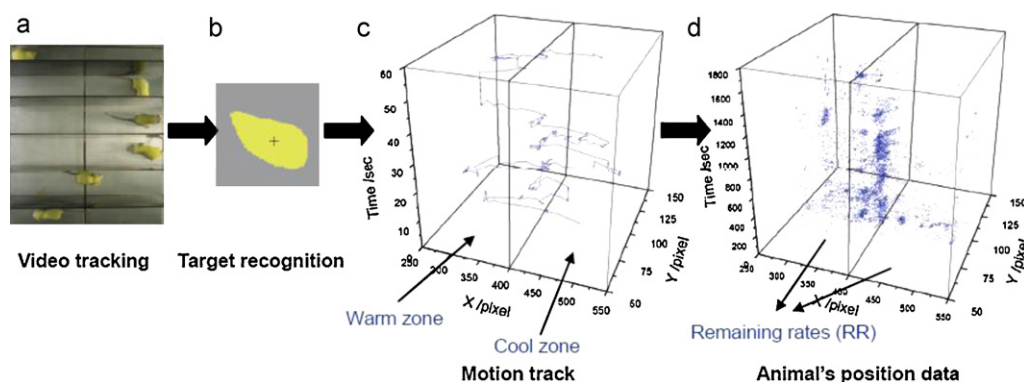


Fig. 2. The schematic diagrams depict the acquisition process of animal thermotropism data. The animal movements are monitored by a video camera, and the recognized signals, the position and the time duration of a mouse staying at the zone, are transformed to electronic data and collected by a computer.

(*Radix aconiti lateralis praeparata*) 20 g kg⁻¹, Ginseng (*Radix ginseng* processed with steam) 35 g kg⁻¹, Coptidis (*Rhizoma coptidis*) 5 g kg⁻¹, Ginger (*Rhizoma zingiberis*) 6 g kg⁻¹, Gypsum (*Gypsum fibrosum*) 3.4 g kg⁻¹ respectively, which were equivalent to human dose of 2.21 g kg⁻¹ Rhubarb, 2.21 g kg⁻¹ Strobal, 3.87 g kg⁻¹ Ginseng, 0.55 g kg⁻¹ Coptidis, 0.66 g kg⁻¹ Ginger, 0.38 g kg⁻¹ Gypsum. The animals were not fast prior to administration of extracts.

All experiments using rodents were performed in accordance with the applicable guidelines and regulations. All mice received humane care in compliance with the institutional animal care guidelines approved by the Ministry of Science and Technology of China.

2.4. Effects of Chinese herbs on thermotropism behavior of normal animals

In order to test if the *cold* and *hot* properties of medicinal herbs can alter the animal thermotropism behavior, we selected several herbal drugs known as inducers of *hot* or *cold* responses and tested them on mice. The experiment was performed on a tri-zone temperature control plate (40 °C, 25 °C and 15 °C, respectively), using herbs Rhubarb (*Radix rhei*), Coptidis (*Rhizoma coptidis*), Gypsum (*Gypsum fibrosum*) as cold response inducers, Strobal (*Radix aconiti lateralis praeparata*), Ginger (*Rhizoma zingiberis*), Ginseng (*Radix ginseng* processed with steam) as hot response inducers.

2.5. Effects of Chinese herbs on thermotropism behavior of model animals

Previous studies have demonstrated that *cold* and *hot* symptoms are the results of the body's energy metabolism (Zhao et al., 2009; Gordon, 1997; Carlisle et al., 1995). Whereas vigorous or over-active physical energy metabolism often shows *hot* physical signs, sluggish or insufficient energy metabolism is often associated with *cold* physical symptoms. Therefore, we induced *cold* and *hot* syndromes in animals by mainly controlling animal food-intake and movement magnitude. To induce *cold* syndrome in animals, mice were fed with limited rations and forced to swim in cold water (10 °C) until nearly drowned, while *hot* syndrome in mice was induced by feeding high-protein diet with unlimited access to food for 7 days. At the same time, the biological markers related to energy metabolism such as adenosine triphosphate enzyme (ATPase), succinate dehydrogenase (SDH), and total anti-oxidant capability (T-AOC) were measured daily in these mice to verify the successful induction of the two types of animal models. We then observed and documented the thermotropism characteristics and the interventional effects of TCM herbal drugs in these animals. The experiment was per-

formed on the bi-zone temperature controlling plates as mentioned above.

2.6. Determination of oxygen consumption

The mice were put in sealed container pre-filled with 10 g of sodica calx on the bottom to absorb CO₂ generated by animals. The consumption of oxygen was measured by the height of water in a U-tube connected with the container (Gao et al., 2004).

2.7. Determination of biochemical indicators

Adenosine triphosphatase (ATPase), succinate deshydrogenase (SDH), superoxide dismutase (SOD), malondialdehyde (MDA) and total anti-oxidation capability (T-AOC) in liver tissue of mice were determined by the aforementioned assay kits. The liver samples of mice sacrificed were removed rapidly into physiological saline maintained at 4 °C to wash the blood on liver surface. Then the liver samples were fresh homogenated to determine the biochemical indicators according to kit instructions.

2.8. Statistical analysis

Values are expressed as mean ± S.D. (standard deviation). Statistical analyses were performed using unpaired *t*-test for quantitative data. *P* < 0.05 was considered significant.

3. Results

3.1. Intervention of herbal drugs on the thermotropism behavior of normal animals

The results showed that normal mice, which were intragastrically administered with Rhubarb, Coptidis or Gypsum, significantly increased the time duration that they stayed in high-temperature zone (*P* < 0.05, Fig. 3a), compared to mice in the control group. On the contrary, when Strobal, Ginger or Ginseng were administered respectively, mice stayed longer and more frequently in the low-temperature zone (*P* < 0.05), short and less frequently in high-temperature zone.

3.2. Thermotropism characteristics of animals with cold and hot syndrome

Based on the above experiments in the normal mice, we tried to induce *cold* or *hot* symptoms in the experimental animals and then intervened with *cold* and *hot* herbal drugs to test the traditional TCM theory to treat cold with hot and to treat hot with cold.

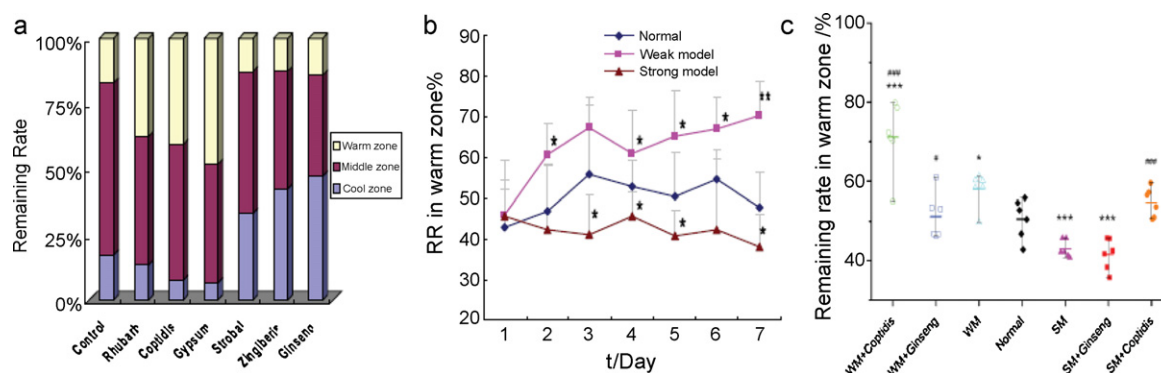


Fig. 3. The effects of cold or hot herbs on the animal thermotropism. (a) The effects of several medicinal herbs on the thermotropism of normal mice. (b) The intervention effects of several medicinal herbs on the thermotropism alteration in cold- or hot-type model animals as described in the main text. * $P < 0.05$, ** $P < 0.01$ compared to animals in the control group; # $P < 0.05$, ## $P < 0.01$ compared to corresponding cold- or hot-type model animals. Colored symbols (circle, triangle, square) represent individual animals in a typical experiment. WM, weak model mice (cold model); SM, strong model mice (hot model). (c) The dynamic trends of the thermotropism in normal, cold-type, and hot-type animals. * $P < 0.05$, ** $P < 0.01$ compared to normal animals.

Table 1

The changes of body weight, body temperature, water intake in mice ($\bar{X} \pm S$).

Group	Body weight (g)	Body temperature ($^{\circ}\text{C}$)	Drinking volume (ml)
Normal	24.97 $\pm$ 3.79	31.65 $\pm$ 1.67	42.33 $\pm$ 2.58
Weak model (WM)	20.86 $\pm$ 0.95 [*]	29.91 $\pm$ 0.73 [*]	28.66 $\pm$ 3.61 ^{**}
Strong model (SM)	26.13 $\pm$ 4.38	30.98 $\pm$ 1.13	47.04 $\pm$ 3.52 [*]
Coptidis + WM	18.18 $\pm$ 2.22 ^{*,#}	29.21 $\pm$ 1.02 [*]	24.30 $\pm$ 1.88 ^{*,#}
Ginseng + WM	23.89 $\pm$ 2.37 [#]	31.28 $\pm$ 0.56 ^{##}	33.25 $\pm$ 3.39 ^{*,#}
Coptidis + SM	28.94 $\pm$ 1.91 [*]	29.01 $\pm$ 0.84 ^{*,##}	50.21 $\pm$ 2.79 ^{**}
Ginseng + SM	25.23 $\pm$ 2.05	30.76 $\pm$ 1.12	52.02 $\pm$ 3.92 ^{*,#}

^{*} $P < 0.05$ compared to animals in the normal group.

^{**} $P < 0.01$ compared to animals in the normal group.

[#] $P < 0.05$ compared to correspondent model animals group.

^{##} $P < 0.01$ compared to correspondent model animals group.

Table 2

The changes of enzyme activities of $\text{Na}^+ - \text{K}^+ - \text{ATPase}$, $\text{Ca}^{2+} - \text{ATPase}$ and $\text{Mg}^{2+} - \text{ATPase}$ ($\bar{X} \pm S$) in mice.

Group	$\text{Na}^+ - \text{K}^+ - \text{ATPase}$ ($\mu\text{mol mg}^{-1} \text{h}^{-1}$)	$\text{Ca}^{2+} - \text{ATPase}$ ($\mu\text{mol mg}^{-1} \text{h}^{-1}$)	$\text{Mg}^{2+} - \text{ATPase}$ ($\mu\text{mol mg}^{-1} \text{h}^{-1}$)
Normal	1.534 $\pm$ 0.191	1.031 $\pm$ 0.072	0.314 $\pm$ 0.032
Weak model (WM)	1.185 $\pm$ 0.080 ^{**}	0.865 $\pm$ 0.198	0.241 $\pm$ 0.041 ^{**}
Strong model (SM)	2.174 $\pm$ 0.214 ^{**}	1.214 $\pm$ 0.113 ^{**}	0.329 $\pm$ 0.037
Coptidis+WM	1.078 $\pm$ 0.197 ^{**}	0.601 $\pm$ 0.101 ^{*,#}	0.186 $\pm$ 0.013 ^{*,#}
Ginseng+WM	1.433 $\pm$ 0.122 ^{##}	1.032 $\pm$ 0.085	0.323 $\pm$ 0.034 ^{##}
Coptidis+SM	1.806 $\pm$ 0.182 ^{*,##}	1.102 $\pm$ 0.325	0.243 $\pm$ 0.011 ^{*,##}
Ginseng+SM	2.021 $\pm$ 0.098 ^{**}	1.393 $\pm$ 0.086 ^{*,#}	0.381 $\pm$ 0.037 ^{*,#}

^{*} $P < 0.05$ compared to animals in the normal group.

^{**} $P < 0.01$ compared to animals in the normal group.

[#] $P < 0.05$ compared to correspondent model animals group.

^{##} $P < 0.01$ compared to correspondent model animals group.

Compared with the normal animals, hunger-and-fatigue induced mice presented traits such as stagnant growth, lower-than-normal body temperature, less motion, disheveled fur and decreased water-intake (Table 1). The observation experiment was

performed on a bi-zone temperature controlled plate (40 $^{\circ}\text{C}$ and 25 $^{\circ}\text{C}$), and the results showed that these animals stayed significantly longer (Fig. 3b) in high temperature zone (40 $^{\circ}\text{C}$) than normal animals, i.e., a behavior of increased thermotropism. In addition,

Table 3

The changes of T-AOC, SOD and oxygen-consumption in mice ($\bar{X} \pm S$).

Group	T-AOC ($\text{U mg}^{-1} \text{protein}$)	SOD ($\text{U mg}^{-1} \text{protein}$)	Oxygen-consumption (ml)
Normal	0.487 $\pm$ 0.052	87.50 $\pm$ 12.85	3.74 $\pm$ 0.85
Weak model (WM)	0.154 $\pm$ 0.053 ^{**}	72.64 $\pm$ 9.04 [*]	1.70 $\pm$ 0.67 ^{**}
Strong model (SM)	0.713 $\pm$ 0.065 ^{**}	136.43 $\pm$ 17.08 ^{**}	4.61 $\pm$ 0.33 [*]
Coptidis + WM	0.113 $\pm$ 0.024 ^{**}	60.81 $\pm$ 5.63 ^{*,#}	0.71 $\pm$ 0.27 ^{*,##}
Ginseng + WM	0.408 $\pm$ 0.041 ^{*,##}	89.45 $\pm$ 8.11 ^{##}	3.29 $\pm$ 0.34 ^{##}
Coptidis + SM	0.563 $\pm$ 0.064 ^{*,##}	120.98 $\pm$ 15.61 ^{**}	5.78 $\pm$ 0.45 ^{*,##}
Ginseng + SM	0.682 $\pm$ 0.072 ^{**}	112.54 $\pm$ 9.02 ^{*,#}	4.16 $\pm$ 0.68

T-AOC, total anti-oxidation capability; SOD, superoxide dismutase.

^{*} $P < 0.05$ compared to animals in the normal group.

^{**} $P < 0.01$ compared to animals in the normal group.

[#] $P < 0.05$ compared to correspondent model animals group.

^{##} $P < 0.01$ compared to correspondent model animals group.

laboratory examinations showed decreased ATPase (Table 2), T-AOC, SOD activity, and oxygen-consumption (Table 3), indicating reduced capacity of energy metabolism and anti-oxidation. On the contrary, the *hot*-syndrome mice induced by nutrient surplus demonstrated antithesis traits as the *cold* ones. These characteristics are basically consistent with the TCM understandings about *cold* and *hot* syndromes in TCM literatures (Székely et al., 2002).

3.3. The interventional effects of medicinal herbs on cold and hot model animals and their thermotropism behavior

Furthermore, we tested the effects of *cold* and *hot* herbal drugs on *cold* and *hot* syndromes in mice, respectively. The results showed that, compared to untreated same type animals, the *cold* herb such as Coptidis could significantly reduce the total residence time of *hot* mice in low temperature zone (25°C), indicating increased thermotropism in these animals (Fig. 3c). As expected, laboratory data of oxygen-consumption and ATPase activity proved that these *cold* herbal drugs could down-regulate the increased metabolic rate of *hot*-syndrome mice treated with *hot* drugs, as stated in traditional TCM literature, *to treat hot with cold*. The *cold* herbal drugs could increase the duration of *cold*-type mice in high temperature zone, transforming *cold*-type animals even *colder*. The interventional effect of *hot* herbal drug such as Ginseng was just the opposite.

4. Discussion

TCM theory states that medicinal herbs present two typical sets with opposite characteristics and these properties can be defined as *cold* or *hot*, based on the responses the herbs induce in living organisms. Judging from the above experimental results, the traditional “*cold*” drug Coptidis could indeed affect the energy metabolism of living bodies and modify animals’ response to environmental temperature, resulting different temperature tropism. It was reported earlier that *cold* or *hot* drugs could change bodily ATPase activity (Ji et al., 2007; Zhuang et al., 1998), and the authors proposed that this phenomenon was due to the drug intervention to energy metabolism (Qi et al., 2001; Zhang et al., 2009). The outcomes of this study were consistent with this proposal. Moreover, we found that the bodily oxygen-consumption and SOD activity, a parameter for bodily energy metabolism and feed-back regulation, altered as well. Altogether, the above results suggested that the *cold* and *hot* properties of medicinal herbs correlated with bodily energy status and its regulation, and, from macro viewpoint, modified the temperature tropism of animals (Sengupta et al., 2004). The *cold* herbs down-regulate, but *hot* herbs up-regulate energy metabolism. These results suggest that *cold* and *hot* properties of the medicinal herbs may be related to the alterations in body energy status and its regulation, hence the animal tropism to different external temperature.

On the other hand, the TCM theory states that there are two opposite body statuses, which express two opposite sets of symptoms and signs, can be defined as *cold* and *hot* syndromes. The individuals with *cold* syndrome have subnormal energy metabolism, as individuals with *hot* syndrome have supernormal energy metabolism. Both the opposite statuses are abnormal but can be corrected by the *cold* and *hot* herbal drugs (Zhou et al., 2009). In *hot* syndrome the abnormally high energy metabolism can be down-regulated to normal with *cold* drugs, *to treat hot with cold*. Similarly, the abnormally low energy metabolism in *cold* syndrome can be elevated to normal level by *hot* drugs, *to treat cold with hot*. Our quantitative results prove that *cold* or *hot* herbal drugs can adjust the level of energy metabolism in animals with *hot* syndrome or *cold* syndrome, respectively. These findings are in accordance

with traditional TCM theory: *treating cold with hot, treating hot with cold*.

To some extent, our results demonstrate that the TCM theory about *cold* or *hot* drugs is the primitive, yet objective and fact-based understandings for drug nature, providing a new perspective to herb-drug study through the Animal Thermotropism Behavior Surveillance System. This System objectively and quantitatively records the animal thermotropism alterations before and after the administration of herbal drugs, with automatic, online, multi-channel (expandable or down-sizable), and multifunctional (temperature control, surveillance, data collection and processing) features.

This is a first report that uses modern approaches of experimental medicine to study the traditional theory about *cold* and *hot* herbal drugs and their interventional effects on animal's search for best-livable temperature. Combined with other biochemical and biophysical analysis, this method may provide a new marker in studying herbal (natural) drugs. The novel methodology may also have applications in other areas of biomedical research, and may impact our fitness and health awareness.

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